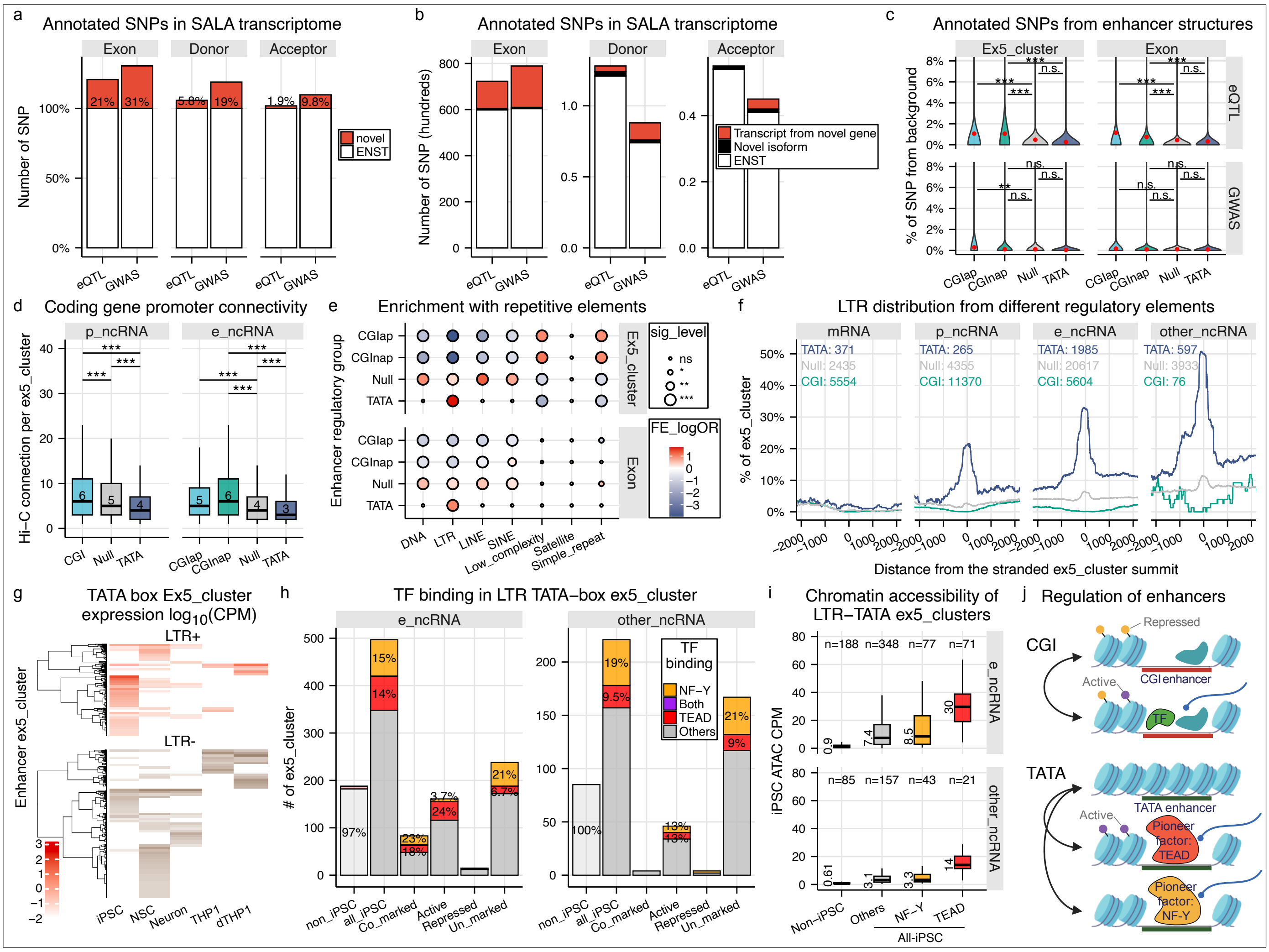

Figure 6 | Genetic architecture, retrotransposon exaptation, and pioneering axes of the non-coding genome.

a, SNPs from eQTL Catalogue and CAUSALdb were intersected with exons of all GENCODE (v39) transcript models and our novel models. SNPs that affect splicing potential from donor sites and acceptor sites were predicted by spliceAI (|delta| ≥ 0.5). A posterior inclusion probability (PIP) ≥ 0.3 was used for eQTLs. The analysis is based on the number of non-redundant SNPs. Percentage increases due to novel annotations are indicated. **b**, SNP intersections as in **a**, grouped by RNA class. **c**, eQTL and GWAS SNPs intersecting e_ncRNAs were grouped by mutually exclusive regulatory elements associated with ex5_clusters. SNP density was normalized to 1 kb of ex5_cluster or transcript sequence. Only SNPs from the “common_all_20180418” panel (NIH) were considered. **d**, Hi-C interactions (5 kb resolution, FDR < 0.01, ≥5 reads) between ex5_clusters linked to ncRNAs and mRNAs. Only clusters with at least one significant interaction are shown. Bins containing multiple regulatory elements (CGI and TATA) were excluded. An interaction observed in any of three cell types was considered positive. **e**, Enrichment of repetitive elements in e_ncRNA ex5_clusters was tested using Fisher’s exact test. A minimum of 6 nt overlap was required for ex5_cluster regions, and 200 nt for merged exon regions. If multiple repetitive elements overlapped, only the longest overlap was retained. **f**, LTR retrotransposon coverage across ex5_cluster summits, grouped into mRNA, e_ncRNA, p_ncRNA, and other_ncRNA and sub-grouped by CGI, Null, and TATA. The total number of ex5_clusters in each group is shown. **g**, Expression profiles of TATA box-associated enhancer ex5_clusters across five cell types. Heatmap is divided by presence or absence of LTR elements. **h**, iPSC-derived binding of NF-Y and TEAD4 on non-redundant LTR-TATA ex5_clusters 501 nt regions. Non-iPSC represents LTR-TATA ex5_clusters that are transcribed only in the other 4 cell types. All-iPSC represents LTR-TATA ex5_clusters that are transcribed in iPSC, which were further divided into 4 groups with distinct histone modification: Co-marked (H3K27ac^{+} & H3K27me3^{+}), Active (H3K27ac^{+}), Repressed (H3K27me3^{+}), Un-marked (H3K27ac^{-} & H3K27me3^{-}). **i**, Chromatin accessibility of iPSC re-counting the single cell-ATAC-seq fragments into the non-redundant LTR-TATA ex5_clusters 501 nt regions. **j**, Schematic illustrating regulatory mechanisms of CGI-derived versus TATA box-derived enhancers. “Active” and “Repressed” refer to the presence of histone modifications H3K27ac and H3K27me3, respectively. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.